



Hugo Folcher

Software Engineer

Mines Saint-Étienne graduate, software engineer since 2021: critical embedded defence systems at Safran, then backend-focused fullstack at Cureety.



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hugo-folcher



hufolcher



Rouen (76000)

PROFESSIONAL EXPERIENCE



SENIOR FULLSTACK SOFTWARE ENGINEER **FULL-TIME · HYBRID | CUREETY | PARIS 75003 | APRIL 2025 - AUGUST 2026**

Oncology remote-monitoring SaaS platform

Patients are followed daily through questionnaires tailored to their treatment. Grew from 10,000 to 50,000 monitored patients during my time, in a team of 15 fullstack developers.

- Led the full rewrite of the Node backend in Rust, a choice driven by my embedded background: from an overloaded service restarted every 30 minutes to a stable backend while load grew fivefold.
- PostgreSQL optimisation: schema, queries, migrations.
- End-to-end tests of the Rust services, with external services simulated.
- Deployment and operations of containerised services on AWS, Jenkins continuous integration.
- Maintenance of the existing LoopbackJS backends.
- Progressive migration of the React/NextJS interfaces to Flutter (Android, iOS, web), plus new features.

Stack: Rust/Actix Web/SQLx, PostgreSQL, REST, NodeJS/LoopbackJS, AWS, Docker, Jenkins, GitLab, OpenTelemetry, Playwright, Dart/Flutter/Riverpod, TypeScript/React, Claude Code

R&D SYSTEM ENGINEER / SOFTWARE DEVELOPMENT **FULL-TIME · ON-SITE | SAFRAN (EX-OROLIA 2S) | SÈVRES 92310 | JAN 2022 - APRIL 2025**

Embedded time & frequency system — Naval Group: maintenance, then redesign

Autonomous high-precision clock distributing time in many electrical and digital formats, in a constrained environment (dedicated electronics, FPGA, software).

- **Design of the new system** — team of 6: 2 software, 2 hardware, 1 physicist, 1 IVVQ
 - ▶ Architecture: business logic, servo loops.
 - ▶ Embedded C/Linux software: scheduling, control of the FPGAs and dedicated boards, UART, TCP/IP over SSL, PostgreSQL.
 - ▶ V-cycle, MISRA C compliant code, SonarQube static analysis.
 - ▶ Python/Qt operator HMI (TCP/IP over SSL, PostgreSQL), screens and procedures designed with the end customer.
 - ▶ Design and code compliant with Naval Group cybersecurity requirements.
 - ▶ Full system simulator (Python/Qt) for test automation and operator training.
 - ▶ Software lead: wrote the technical documentation and defended design choices in reviews with Naval Group experts.
 - ▶ Technical documentation and reports generated automatically (Typst).
- **Maintenance of the legacy system, untouched for years** — paired with a hardware engineer
 - ▶ Took over a C/Linux computer with no documentation and no remaining expertise: code audit and reverse engineering of its behaviour.
 - ▶ Bug fixing of the computer's software.
 - ▶ Simulation engine for the electronic and FPGA environment (Python/Qt, UART) to test and validate every software release without the target hardware.

Stack: C, Python, PyQt/PySide, PostgreSQL, Docker, VirtualBox, GitLab

Resilient navigation unit (NavKite)

Inertial navigation unit coupled with a GNSS receiver that detects malicious signals: from prototype to a Safran Electronics & Defense catalogue product. Paired with a hardware engineer.

- Python backend: acquisition and control of the inertial unit, protection against malicious signals, API for the web interface.
- Proof of concept within weeks, then continuous improvement and field demonstrations.
- Industrialisation: packaging and custom Linux OS (Buildroot).

Stack: Python/Gunicorn/Socketio, React, Docker, Buildroot

GNSS simulation for a defence application (Skydel)

Drone spoofing prototype broadcasting modified GNSS signals. Solo project.

- Adapted a civil simulator to critical defence use (C++17, Qt, encryption).
- C++ backend supervising and controlling the whole system (simulator, hardware), exposed as an API.
- Integration with other systems and customer demonstrations.

Stack: C++17, C++20, Qt, GitLab

Atomic reference time card (ARTCARD)

Open-source card co-developed with Meta for datacenters: FPGA, u-blox GNSS receiver, MRO50 miniature atomic clock. Team of 2 software engineers and 1 physicist, working with Meta's team.

- Clock disciplining on GNSS in C, open-source development and support (github.com/Orolia2s/oscillator).
- Contribution to the open-source Linux driver, since merged into the kernel (github.com/Time-Appliances-Project/Time-Card); internal fork for prototypes.
- Programming (FTDI, FPGA, EEPROM) and test bench, with an operator HMI for production at the EMS AsteelFlash.

Stack: C, Python, PyQt/PySide

EDUCATION



ÉCOLE NATIONALE SUPÉRIEURE DES MINES DE SAINT-ÉTIENNE | 2018 - 2021

Campus Georges Charpak Provence - Gardanne, 13120

- ISMIN Engineering Program (Microelectronics & Computer Science)
- Specialization in Microchip Design
- Technology Focus: Information Technologies & Supply Chain

CLASSE PRÉPARATOIRE AUX GRANDES ÉCOLES (MPSI-MP*) | 2015 - 2018

Lycée Montaigne - Bordeaux, 33000

- MPSI-MP* - Computer Science Option / 2015 - 2017

Lycée Godefroy de Bouillon - Clermont-Ferrand, 63000

- MP - Computer Science Option / 2017 - 2018

SCIENTIFIC BACCALAUREATE WITH HIGH HONORS | 2015

Lycée Charles Gide - Uzès, 30700

SKILLS



Languages: C, C++, Rust, Python, SQL, Bash, TypeScript, JavaScript, Dart

Frameworks / Tools: Linux/UNIX, POSIX, TCP/IP (SSL/TLS), Linux drivers, UART, FPGA, MISRA C, SonarQube, Git, GitLab, Github, Jenkins, GitHub Actions, Docker, AWS, GCP, Yocto, Buildroot, GDB, Valgrind, Wireshark, Qt (C++), PySide/PyQt, PostgreSQL, SQLx, Actix Web, SQLAlchemy, Grafana, OpenTelemetry, Playwright, Typst, NodeJS, Flutter, React, Claude Code

LANGUAGES



FRENCH (Native)

ENGLISH (Professional)

SPANISH (Conversational)

INTERESTS



HOME AUTOMATION & IOT (Home Assistant, Zigbee, Thread, ESP32, TrueNAS NAS, networking)

SPORT (running, hiking, climbing, sailing)